VTES Card Scanner

User manual



Real-time VTES card identification for OBS Studio

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La Garra Cifrada

github.com/ChicoCifrado/VTES-for-OBS

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> **How to read this manual**
> Sections 1–4 will get you running in a few minutes. Sections 5–8 describe what you see on screen and everyday use (including director mode). Section 9 explains the automatic game state and section 10 gathers the typical problems with their solutions.

1 · What is VTES Card Scanner?

VTES Card Scanner is a plugin (filter) for OBS Studio that identifies Vampire: The Eternal Struggle cards in real time as they appear in front of your camera, and builds a broadcast-ready stream with them: your camera inside a decorative frame, the shown card on the ankh and game information on screen.

In practice, it gives you:

- Automatic identification: YOLO OBB detector + visual embeddings as the main path even on small cards + type classifier (14 card types). OCR on the name is only a backup for close-ups from 200 px high; the final decision comes from the embedding.
- Broadcast overlay: your camera inside the ankh design, the last approved card shown whole over the symbol, and the rulings drawn on top of the card.
- Control web app (phone or PC): card search, pool control, live detections, game state and the pending queue.
- Automatic game state: per-player zones, locked/unlocked cards, influence, eliminated players and an event log.

- Director mode: detections are not shown automatically; YOU decide what gets drawn by reviewing the pending queue.

> **Fan project**
> VTES Card Scanner is a community project, not related to Paradox Interactive. Vampire: The Eternal Struggle is a trademark of Paradox Interactive. The plugin is distributed under the GNU GPL v2 license.

2 · Requirements

- OBS Studio 30 or newer, on Windows 10/11 64-bit.
- Camera: a USB webcam. The better the optics and focus, the better the embedding works; it is the main path, while OCR only runs on large cards. The camera must be able to frame the cards at a comfortable distance.
- GPU: NVIDIA with CUDA (recommended). Alternative: any GPU via DirectML. CPU mode available (slow; testing only).
- Even lighting over the table, with no direct glare or abrupt light changes.
- Optional: a phone on the same WiFi network to control the game during the stream.
- The installer bundles the AI models, OpenCV, ONNX Runtime and Tesseract OCR 5.x; no extra installs needed.

3 · Installation

Option A — Installer (recommended)

1. Download VTES-for-OBS.exe from the project's Releases page.
2. Run it and follow the wizard: the installer locates OBS automatically from the Windows registry (default: C:\Program Files\obs-studio).
3. Open OBS and add the filter to your camera (section 4).

To uninstall: close OBS and use «Add or remove programs» → VTES Card Scanner OBS Plugin → Uninstall (or run uninstall-vtes-card-scanner.exe from the plugin folder).

Option B — Manual copy (zip)

1. Download the release .zip and extract it.
2. Copy its contents into your OBS folder (usually C:\Program Files\obs-studio), keeping the folder structure (obs-plugins\64bit\ plus the plugin data).
3. Restart OBS.

Building from source (developers)

Clone the repository and use the vtes-grimoire.ps1 script (build / deploy). The AI models are not in the repository; see the README to place them in data\obs-plugins\vtes-card-scanner\.

Notice when adding the filter
The first time you add the filter to a source, OBS shows a notice with the control web address (http://localhost:8080) and quick tips: enable «Always active» for continuous detection and adjust the interval if you want to lighten the load.

4 · Getting started

1. Add a «Video Capture Device» source to your scene, with your camera pointing at the table.
2. With the source selected: right-click → Filters → + → «VTES Card Detector (OBB)».
3. In the filter properties choose the inference device: GPU (CUDA) if you have NVIDIA; GPU (DirectML) for any other GPU; CPU as a last resort.
4. Enable «Show bounding boxes» while testing: each detected card appears with its box and name. You can leave it on or turn it off later; the broadcast overlay does not depend on it.
5. Hold a card in front of the camera: it should be identified in under a second and appear on the web. If director mode is off, it will also be drawn straight onto the ankh.
6. Open the control web with the «Open card search» button or by visiting http://localhost:8080.

Camera tips
Frame the cards as large as possible and inside their player's zone. Avoid shadows and glare on the card: the embedding uses the whole image and OCR only runs when the card is a close-up at least 200 px high. If your table is at a steep angle, raise the camera a little.

5 · The broadcast overlay (the ankh design)

The filter's output is the final broadcast image: your camera covers the decorative design's window («cover» scale: it fills the window with no black bars, cropping the excess) and the design is drawn on top with real transparency. Zoom (100–300%) and position let you choose the framing; the offset is clamped automatically so the camera never leaves edges. With «Full-bleed camera» the camera also fills the side bands and the design is drawn on top. The design is 3:2 (1536×1024), so a 16:9 scene without full-bleed will have bands at the sides: that is part of the style. Framing only affects the visual composition; detection and the embedding still use the full frame, and OCR only runs on large cards.



Overlay anatomy: the real design with the camera in its window.

- Left panel: one row per player (up to 5) with their name and pool.
- Ankh: the last card sent to the overlay is shown whole over the symbol.
- Card icons: a row under the shown card with its type, clan or creed and disciplines or virtues. They are the official KRCP symbols, deduplicated and fitted to the card width.
- Rulings: the rules text of the shown card is drawn on top of it; its height is adjusted with «Rulings scroll».
- Camera window: your video feed, cover-fitted and adjustable with zoom and position.
- Full-bleed camera: the camera also fills the side bands; the design is drawn on top.
- Names and pool: drawn with the selected font (Cinzel, Grenze Gotisch or Classic).
- Big card: takes half the screen; meant for streams watched on a phone (enable it in the properties).
- Show zones: draws the rectangles of each player's zones (minions and crypt). Useful for calibration; turn it off to broadcast.

6 · Filter settings

All options live in the filter properties (right-click the source → Filters → VTES Card Detector). Defaults in parentheses.

General

Language
> Spanish or English. Changes the plugin and web texts.

Show bounding boxes (on)
> Draws detection boxes and names over the video. Essential for calibration; you can turn it off afterwards.

Always active (on)
> Keeps detection running even when «Show bounding boxes» is off. Leave it on while streaming.

Big card (off)
> The shown card takes half the screen. For tournaments and streams watched on a phone.

Rulings scroll (0 %)
> Moves the rulings block up or down over the shown card (0–100 %). 0 is the base position.

Font and framing

Player name font (Cinzel)
> Cinzel (Roman) is the default; «Grenze Gotisch (gothic)» and «Classic (OpenCV)» are the alternatives. If a font is missing, the classic one is used. Affects the panel's names and pool.

Camera zoom (100%)
> 100 to 300%. 100% is the minimum crop that covers the window; raise it to zoom in. In full-bleed mode you need more than 100% to be able to move the framing.

Camera horizontal / vertical position (0)
> Moves the framing. The plugin clamps the offset so the camera keeps covering the window (or the whole frame, with full-bleed).

Full-bleed camera (off)
> The camera fills the whole frame, including the side bands of the 3:2 design; the design is drawn on top. Useful for wider shots.

Detection

Confidence threshold (0.50)
> From 0 to 1. Raise it if false detections appear; lower it if cards are hard to detect in poor light.

Detection mode (AI Model — ONNX)
> «AI Model (ONNX)» is the normal mode. «Shape (contour)» is a backup detector without AI for extreme cases.

Inference device (GPU CUDA on Windows)
> CPU, GPU (DirectML) or GPU (CUDA). CUDA is fastest with NVIDIA; DirectML works with any GPU; CPU is testing only.

Detection interval (0 s)

Minimum time between detections, 0 to 5 seconds. 0 = continuous. Raise it (0.5–1 s) if the PC struggles or you use CPU.

Type classifier (on)

Vision network that recognises the 14 card types (Action, Master, Combat…). Improves filtering and the web info.

OCR — read card names with Tesseract (on)

Reads the printed name and matches it against the card database. It only runs when the card is at least 200 px high in the frame and the embedding is not already confident; it is not used on small cards. If the embedding is torn between two cards, the decision goes to the Pending queue.

> **Small cards**
> The detector and embedding are designed for real cards at reduced size. The smaller the card, the more focus and light matter; OCR is not forced below 200 px. If two candidates appear, choose in Pending instead of trusting a text read.

NIS upscaling (off, requires CUDA)

Sharpens small text before OCR. Better reading at some GPU cost.

CUDA Oval/Rect classifier (on)

Tells cards apart from other rectangles (sleeves, phones, the playmat): fewer false positives. Requires CUDA.

Director mode

«Director mode: detections are not shown automatically…» (on)

When on, every detection goes to the Pending queue and only what you tap gets drawn (section 8). When off, the last identified card is shown automatically (classic mode).

Players

Player count (5)

4 or 5 players. The overlay panel has 5 rows.

Show player zones (on)

Draws the zone rectangles over the video (calibration).

«Player N» group

Name, Initial pool (30) and «In game». Plus the geometry of their two zones —minions and crypt— with X/Y/Width/Height as normalised values (0–1 relative to the source size) and angle (−180° to 180°). A card is assigned to the player whose zone contains its centre.

Open card search (button)

Opens the control web in your browser; starts the server if it is not running.

7 · The control web app

The plugin starts a web server on port 8080 when the filter is added. From there you control everything during the game, from the PC or the phone.

- On the PC: http://localhost:8080
- On the phone (same WiFi): http://PC-IP:8080 — the first time Windows will ask to allow OBS on the private network; accept the prompt.

The web uses the same official icon fonts as the overlay: every card shows its type, clan or creed and disciplines or virtues. The fonts are served from the plugin's own /fonts/ path; no connection is needed.

The top bar links every page: Search · Pool · Detections · Game · Pending.

Search (/)

> Search across the card database. Accepts name (Spanish, English or French), type:, clan:, disc: (discipline abbreviation), cap: (exact capacity), cap>N, group: and free text, all combinable. Shows up to 50 results with the official type, clan and discipline icons; open a card to send it to the overlay with one click.

Pool (/pool)

> Pool control per player with buttons. The pool cannot be read from the camera (bleeds, costs), so it is adjusted by hand here or from Game.

Detections (/detections)

> Latest live detections with image, name, time and icons. One click on any of them draws it on the ankh instantly.

Game (/game)

> Game state: per player, the pool, how many cards are locked or unlocked, detected minions and crypt cards. Below, the Log with recent events (locks, influence, pool, ousted) and their age.

Pending (/pending, alias /pendientes)

> The director mode queue: each pending card shows its icons and here you approve what gets shown on the overlay (section 8).

8 · Director mode (visual check)

It is the recommended way to broadcast: the plugin detects and identifies, but draws nothing on its own. Detections pile up on the web and you approve which one is shown. That way a doubtful detection never goes on stream... and every choice also becomes data to improve future versions.

Workflow

1. With «Director mode» on (it is by default), show cards to the camera as usual.
2. Open Pending on the phone or PC: every detected card appears with its image, its evidence crop and its confidence level. Nothing is drawn on the overlay yet.
3. Tap the right card: it is drawn instantly over the ankh and the whole queue is cleared (the rest disappear).
4. If none of them is usable, press «Discard all».
5. To clear the ankh whenever you want, press «Remove from overlay».

- Per-card dedupe: if the same card is detected again, it refreshes in the queue instead of duplicating.
- If two cards are nearly tied (for example, in a blurry capture), the queue shows both candidates so you can pick the right one; the plugin draws neither of them automatically.
- The queue holds up to 24 pending cards; ones not seen again expire after 120 seconds.
- The view is shared between devices: phone and PC see the same queue; approve from wherever is handiest.
- With the mode off you return to the classic behaviour: the last identified card is shown automatically.

Training data

Every director mode choice generates data on your own machine (it never leaves your equipment). In the plugin's config folder you will find the history and the crops of each decision:

```
%AppData%\obs-studio\plugin_config\vtes-card-scanner\feedback\
events.jsonl      history of approvals and discards
events\<id>\      chosen image + candidates of each event
```

You can delete that folder whenever you want; if you keep it, it helps improve recognition in future versions of the plugin.

9 · Automatic game state

With the zones configured, the plugin works out the table situation from what it sees.

Player zones

- Each player has two rectangles: minions (top by default) and crypt (bottom), both with adjustable angle.
- Adjust X/Y/Width/Height as values from 0 to 1 relative to your source size; with «Show player zones» enabled you will see them drawn on screen.
- A card belongs to the player whose zone contains its centre; if it falls in none, it is «outside» and does not count for the state.

Locked cards (lock)

- Only cards in the minion zone are evaluated: crypt cards are not tapped.
- The tilt is measured against the zone angle: ≈90° = locked; back in place = unlocked. Evaluation is continuous: a card can lock or unlock at any point in the game.
- The state shows on /game (locked/unlocked per player) and in the log.

Influence (crypt → minions)

When a card moves from the crypt zone to the minions zone (influence completed on the real table), the change is recorded in the Log as «influenced».

Pool and elimination

- The pool is controlled from the web: the plugin cannot read blood counters or pool through the camera.
- If a pool reaches 0, the player is marked as ousted, with a label and a dimmed look. If you raise it again (say, an accidental tap), it reactivates on its own.

Log

On /game you get the latest events with their age: locks and unlocks, crypt-to-minions moves, pool changes and eliminations/reactivations.

10 · Troubleshooting

No card is detected

Check the OBS Log (Tools → Log files) that the models load when the filter starts. Improve the lighting and fill more of the frame with the card. If needed, lower the confidence threshold to 0.35–0.45. Confirm «Always active» is checked.

False positives (it detects things that are not cards)

Raise the confidence threshold; enable the «CUDA Oval/Rect classifier» and «NIS upscaling» if you have NVIDIA; slightly raise the detection interval.

It is slow or the PC suffers

Switch from CPU to GPU (DirectML or CUDA); raise the detection interval to 0.5–1 s; disable NIS upscaling if it was on.

The web does not open on the PC

The server starts when the filter is added to some source: check that. If another program uses port 8080, the OBS Log will show a warning.

The phone cannot reach the web

Use the PC's IP (not «localhost») and make sure you are on the same WiFi. The first time, allow OBS in the Windows Firewall for private networks.

In director mode nothing shows on the ankh

That is expected: nothing is drawn automatically. Approve cards from the Pending page.

I tap a pending card and nothing is drawn

That detection ended up without an associated image. Show the card to the camera again and use the new pending entry, or draw it from Detections.

Lock detection is inverted

Check that player's minion zone angle in the properties: it must match the real orientation of the table.

OCR mixes up card names

Improve the light and avoid glare on the card. Image embedding is the main path and fixes most doubtful cases; OCR only runs on cards at least 200 px high.

A small card is not identified

Move it closer or improve focus: the embedding needs a sharp card. OCR does not run below 200 px; if the embedding returns two candidates, they appear in Pending for a one-click choice.

Where do I check errors?

Tools → Log files in OBS. The [Game], [Director], [WebServer] and [OCR] tags follow the plugin flow.

11 · Privacy and data

All processing is local: the camera, the detections and the web run on your machine, and the plugin sends nothing to the internet. The only data it generates are its configuration and the optional director mode feedback, in:

```
%AppData%\obs-studio\plugin_config\vtes-card-scanner\
  (config, copied card database and optional feedback\ folder)
```

You can delete that folder whenever you want.

12 · Credits and license

- Project: VTES Card Scanner — La Garra Cifrada.
- Repository and issues: github.com/ChicoCifrado/VTES-for-OBS
- License: GNU GPL v2.0 (see the LICENSE file in the repository).
- Vampire: The Eternal Struggle is a trademark of Paradox Interactive. This project is unofficial fan-made content.

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